Chapter 5 — Feature engineering and transforms

Many models need more than scaled raw columns

Learning objectives

- Explain equal-width versus frequency binning, construct ratio and interaction features
- Choose log or polynomial expansions when skew or nonlinearity demands them

Binning numerical data

- Binning groups continuous values into intervals
- Example 5.51 uses equal-width age bins
- Binning reduces noise and handles nonlinear effects but loses granularity

Adding new features

- Example 5.53 adds a squared term
- Domain ratios often outperform raw stacks of columns
- Avoid generating features that duplicate identifiers or leak targets

Example 5.55 — Engineer price per square foot

- Example 5.55 — hands-on module
- The classic ratio feature
- Explore the chapter example module
- View files: `modules/chapter5/example55/`

Dimensionality reduction and selection

- High-dimensional sparse tables benefit from principal components
- Selection reduces overfitting and training time
- Keep interpretability requirements in mind for regulated domains

Log and polynomial transforms

- Log transforms compress right tails, Example 5.26 on income
- Polynomial expansions, Example 5.56, let linear models capture curvature
- Apply transforms before splitting or fit parameters on training folds only
- Inverse transforms may be needed for interpretation

Takeaways

- Engineering encodes domain knowledge
- Binning and transforms trade resolution for stability
- Document every derived column for reproducibility and explainability

Next

- Complete the quiz for this part
- Continue to Python tools that implement these steps at scale